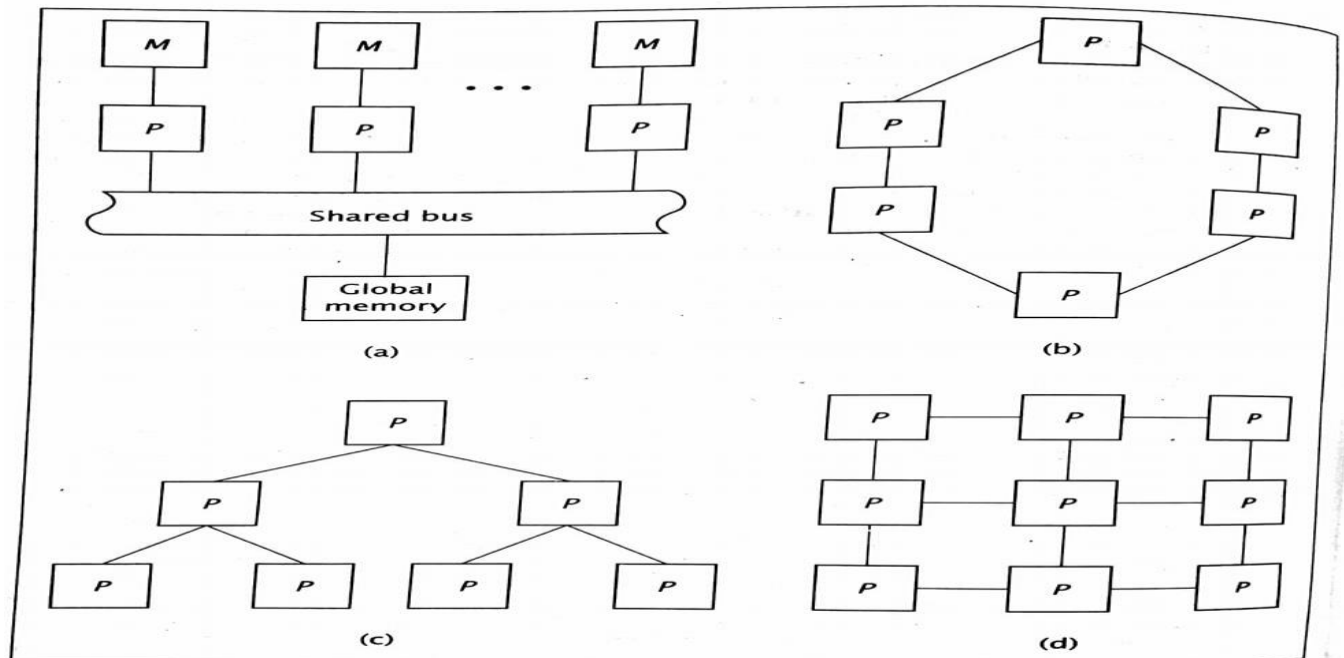
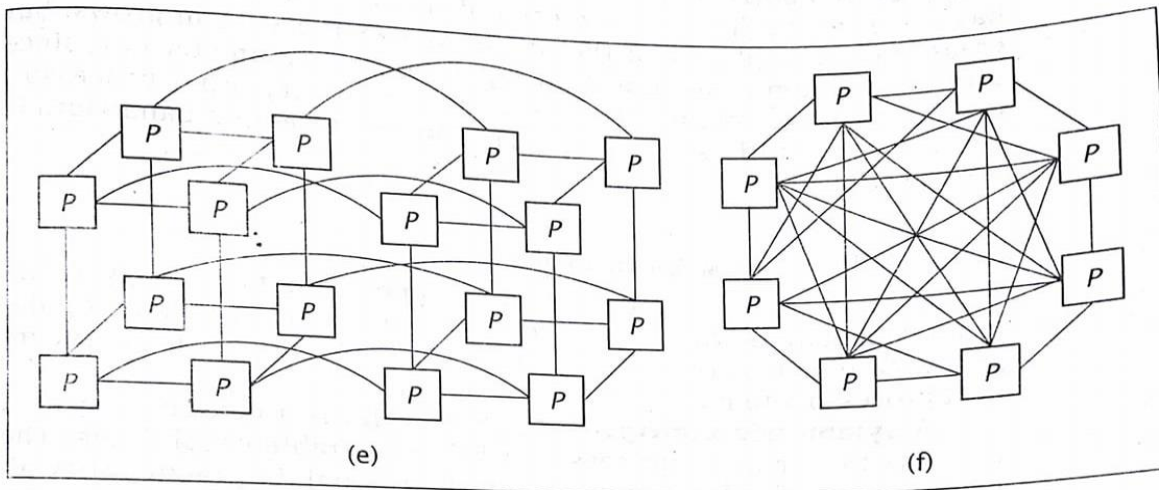


System Topologies/Interconnection Structures: (How multiple processors are connected with memory and i/o devices?)

- Pattern of connections between the processors in a multiprocessor system.
 - **Diameter**- maximum distance between 2 processors in the system.
 - **Bandwidth**- capacity of communication link(L) multiplied by the number of such links in the system.
- A. Shared Bus
B. Ring Topology
C. Tree Topology
D. Mesh Topology
E. Hypercube Topology
F. Completely connected Topology

MIMD system topologies: (a) shared bus, (b) ring, (c) tree, (d) mesh, (e) hypercube, and (f) completely connected





■ A. **Shared Bus**

- => Processor communicates each other via bus.
- => Bus can handle only one data at time.
- => Processors directly communicate with their own local memory to limit traffic on bus.
- => Easy to expand.
- => Diameter = 1 and Bandwidth = 1. L

■ B. **Ring Topology**

- => Uses direct dedicated connections between processors unlike shared bus.
- => All communication line to be active simultaneously.
- => Data travels through several processors to reach final destination.
- => Diameter = $n/2$ (largest integer less than or equal to $n/2$)
- => Bandwidth = $n \cdot L$

■ C. **Tree Topology**

- => Direct connections between processors; each processor has three connections.
- => Primary advantage is its low diameter.
- => Diameter = $2(\lg n)$; where $\lg n$ is the base 2 logarithm of the number of processors.
- => Bandwidth = $(n - 1) \cdot L$

■ D. **Mesh Topology**

- => Every processor connects to the processors above and below it, and to its right and left.
- => Without wraparound connection (we can wrap and overlap from left to right and top to bottom, eg as in the figure)
 - Diameter = $2\sqrt{n}$ and Bandwidth = $(2n - 2\sqrt{n}) \cdot L$
- => With wraparound condition
 - Diameter = $\sqrt{n}$ and Bandwidth = $2\sqrt{n} \cdot L$

■ E. **Hypercube**

- => Is a multidimensional mesh.
- => It has n processors, each with $(\lg n)$ connections.
- => Diameter = $(\lg n)$ and Bandwidth = $(n/2) \cdot (\lg n) \cdot L$

► F. **Completely Connected Topology**

=> Each processor has $(n-1)$ connections.

=> Increases the complexity of processor.

=> Diameter = 1 and Bandwidth = $(n/2) \cdot (n - 1) \cdot L$